

# AI for Communication and Marketing

Apr 22, 2026 – Lab lecture #7: Repurchase models



# Hands-on Session: Your repurchase model challenge

Same DS as lecture number 3

Dataset Information

Additional Information

This is a transactional data set which contains all the transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based and registered non-store online retail.The company mainly sells unique all-occasion gifts. Many customers of the company are wholesalers.

| Variables Table |         |             |                                                                                                                                   |
|-----------------|---------|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Variable Name   | Role    | Type        | Description                                                                                                                       |
| InvoiceNo       | ID      | Categorical | a 6-digit integral number uniquely assigned to each transaction. If this code starts with letter 'c', it indicates a cancellation |
| StockCode       | ID      | Categorical | a 5-digit integral number uniquely assigned to each distinct product                                                              |
| Description     | Feature | Categorical | product name                                                                                                                      |
| Quantity        | Feature | Integer     | the quantities of each product (item) per transaction                                                                             |
| InvoiceDate     | Feature | Date        | the day and time when each transaction was generated                                                                              |
| UnitPrice       | Feature | Continuous  | product price per unit                                                                                                            |
| CustomerID      | Feature | Categorical | a 5-digit integral number uniquely assigned to each customer                                                                      |
| Country         | Feature | Categorical | the name of the country where each customer resides                                                                               |

# Hands-on Session: Your repurchase model challenge

01

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## EDA & quality check (you can reuse what you did in lecture 3)

Load the data and try to answer three key questions:

1. What does the data look like? (shape, first rows, summary statistics)
2. Which columns have missing values and how many?
3. Are there any obvious outliers or impossible values?

Handle outliers and missing values.

03

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## Modeling

Train the model/models and evaluate it using Accuracy, Precision, Recall, F1-score, AUC/ROC score and Lift.

Calculate feature importance.

02

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## Defining the Target Variable & Feature Engineering

- Split the dataset into a 'Feature Window' and a 'Target Window' based on .. ?
- Create a binary target variable.
- **Feature engineering:** based on the Feature Window data, generate a customer-level dataframe with the features you think could be relevant.

04

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## Hyperparameters tuning

Use Randomized Search/Grid Search to tune the hyperparameters and compare the results with the baseline model's.